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Palaeomagnetism and palaeogeography of Mongolia from the Carboniferous to the Cretaceous—final report

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ABSTRACT

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The present study summarizes the results of palaeomagnetic investigations of Carboniferous to Cretaceous rocks with respect to global-tectonic interpretations of Eurasia. Normal and reverse directions of stable remanence components were found for the Carboniferous to Cretaceous rocks within single outcrops, sometimes even within individual samples. Triassic, Permian and Carboniferous pole positions for Mongolia are strikingly different from those of the Siberian platform. The conclusion about the different palaeogeographical development of Mongolia with respect to the Siberian platform is well seen from differences in palaeolatitudes (up to 50°) and palaeorotations (up to 150°). The apparent polar wander paths for Mongolia and the north China block are almost identical since the Carboniferous. This way, palaeomagnetic and palaeogeographical data are contributing to the global-tectonics model of the regions under study. During the Carboniferous, Permian and Triassic, the regions of Mongolia investigated were not part of Eurasia. Together with the north China block, they show a palaeogeographical affinity with the Pacific Plate.

1. Introduction

The vast area of China, Mongolia and Siberia is a tectonically very important and complicated part of Eurasia. Despite the facts that are known, not all problems have been solved. Was the region of Mongolia really part of the Siberian platform in geological history and thus also of Laurasia? When exactly did the various tectonic plates of China and Mongolia become part of the northern continent? In order to answer these questions, and to compile palaeogeographic maps, we have to continue palaeomagnetic and palaeogeographic studies. The present paper furnishes more information on the basis of palaeomagnetic data on the one hand, and essentially new views of the global tectonics of central Asia, on the other.

2. Notes on tectonic evolution

In the present structure of the Asian continent five continental plates (blocks) are known: the Siberian, Chinese–Korean, South Chinese (blocks), and Hindustani and Arabian plates. The Siberian Plate is separated from the East European (Russian) Plate and Chinese–Korean block by the Ural–Okhotsk (Ural–Mongolian) mobile belt, which has a meridional fault in the northwestern part and a parallel fault in the southeastern part, known as the Central Asian zone (Jansin et al., 1984).

The tectonic structure of Mongolia is a symmetric configuration of sutures and subduction zones. The northern mobile belt is wider and vaster than the southern one, and is also referred

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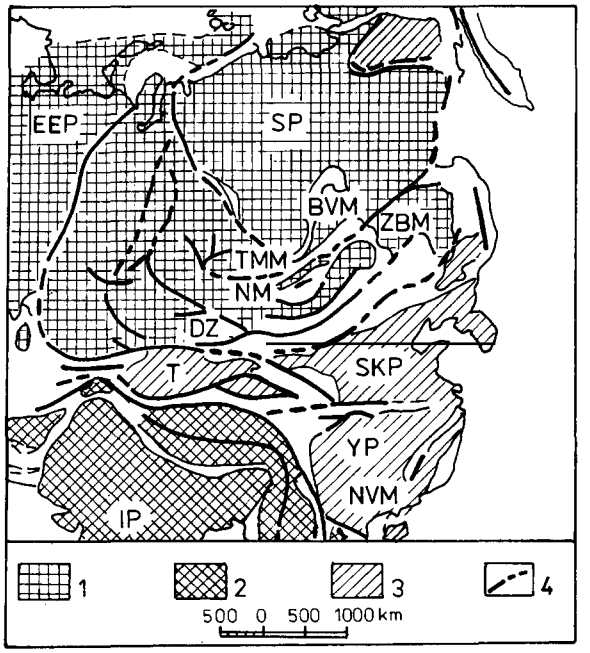


Fig. 1. Continental blocks and ophiolite zones of Asia (after Jansin et al., 1984). 1, the Laurasian Plate; 2, the Gondwana Plate; 3, the 'Pacific' Plate; 4, boundary of ophiolite and suture zones from the Rifean to the Cretaceous. Old plates: EEP, East European; IP, Hindustani; SKP, Chinese–Korean block; SP, Siberian; YP, Yangtze block. Massifs: BVM, Bajkal–Vitim; DZ, Dzhungaria; NM, north Mongolian; NVM, North Vietnam; T, Tarim; TMM, Tuva–Mongolian; ZBM, Zeya–Bureya.

to as multicyclic. The beginning of the formation of structures can be traced as far back as the Proterozoic.

The Chinese–Korean block comprises north China and a greater part of the territory of Korea. In the north it is bounded by the Mongolian geosyncline and in the south by the Qinling geosyncline, which separates it from the Yangtze block. The Mongolian geosyncline, which runs in the north along the Chinese–Korean and Tarim blocks is, together with the Ural and Okhotsk fold arc, part of the large geosynclinal Ural–Mongolian–Okhotsk mobile zone with a lot of oceanic-type sediments.

Continental blocks and ophiolite zones of Asia (after Jansin et al., 1984) are presented in Fig. 1. We can see, as well as the Laurasian Plate, the Gondwana Plate, and the 'Pacific' Plate, smaller blocks lying partly between belts in the mobile zone. Some of them are important for this palaeomagnetic research—the Tuva–Mongolian, Zeya–Bureya, and the north Mongolian massifs. On the basis of tectonic development and the formation of geological structures, some authors, for example, Zajcev and Luvsandanzan (1977), Zonenshain (1977), Kravchinskij (1979), and Jansin et al. (1984), include the blocks of Mongolia in the Siberian Plate.

3. Palaeomagnetic investigations

Palaeomagnetic investigations of Carboniferous to Cretaceous rocks pointed out in the first instance (Pruner, 1987) that it was not sufficient to apply demagnetization procedures to pilot samples only, but that stepwise demagnetization of entire sets of samples was required. In laboratory stability tests we employed alternating fields (AF) up to a maximum amplitude of 100 mT by means of the Schonstedt GSD-1 apparatus, and thermal demagnetization was effected using the MAVACS apparatus (Příhoda et al., 1989).

The rock samples under investigation were measured by highly sensitive spinner magnetometers of the JR-4 type (Jelínek, 1966). After each heating step the volume magnetic susceptibility was measured in order to identify possible phase changes of minerals in the rocks. Measurements were made with the use of the KLY-2 induction bridge (Jelínek, 1973).

Of the total number of 550 samples investigated from 35 localities, Table 1 showing the mean values of palaeomagnetic directions presents statistical processing of 310 samples from 18 localities. Some rock samples were remagnetized or exhibited unstable magnetization components. In the paper (Pruner, 1987) describing preliminary data, altogether 177 samples from nine localities were used for palaeogeographical interpretation.

4. Upper Carboniferous

Examples of effective demagnetization by the AF field (12–30 mT) are Carboniferous tuffites collected from three exposures in the Gobi

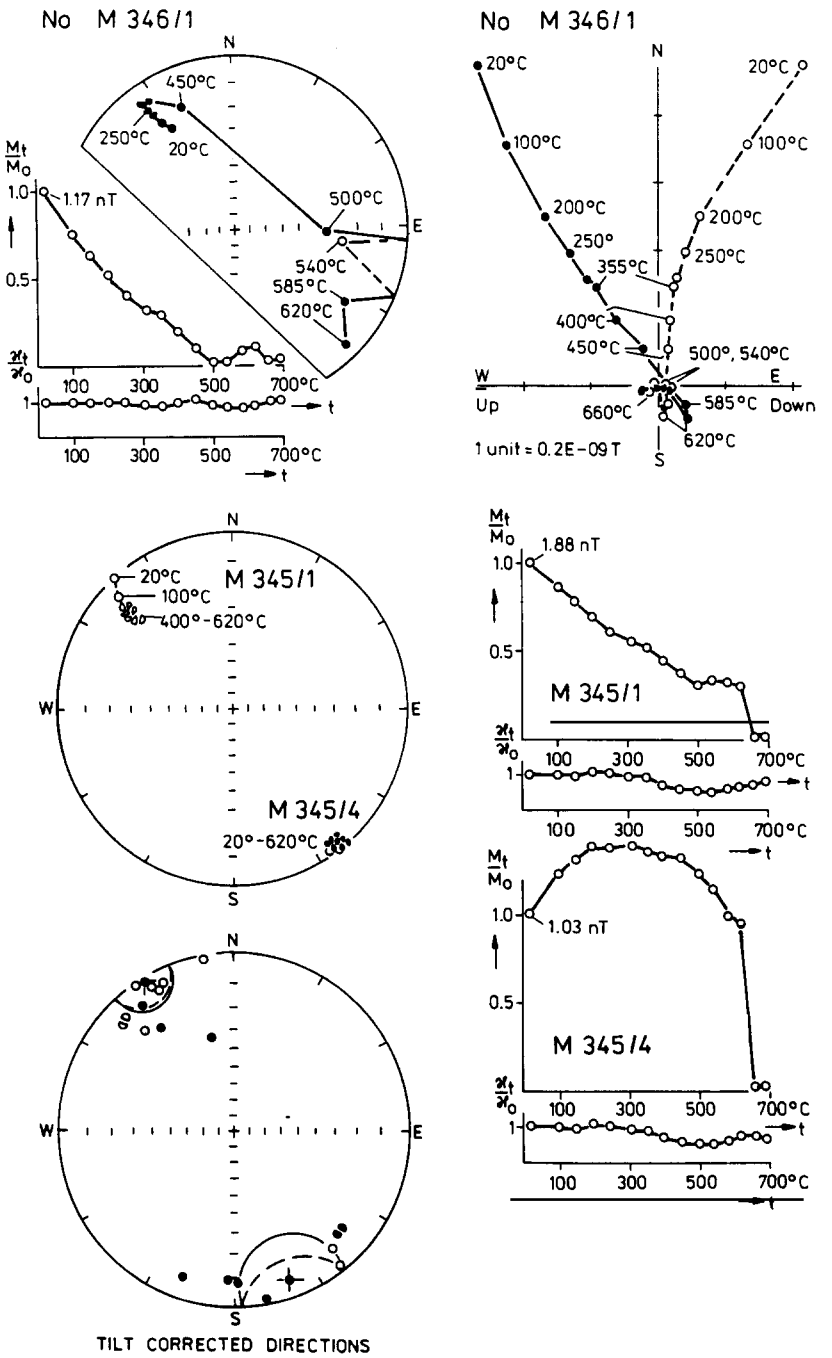


Fig. 2. Results of different types of thermal demagnetization of Upper Carboniferous tuffites from the Urtuchuraj locality, site 29. The figure shows normalized graphs of M_t / M_0 and χ_t / χ_0 dependent on temperature and stereographic projection of remanent magnetization on to lower (upper) hemisphere by full (open) circle. The circles of confidence were calculated according to Fisher (1953) at the 95% probability level. In the Zijderveld diagram the full (open) circles denote projection on to horizontal (vertical) planes.

TABLE 1

Review of collected samples in Mongolia

Sample nos.	No.	Locality	Rock type	Mean palaeomagnetic directions		α 95	K	n	Notes, stability tests	
				D(°)	I(°)					
Carboniferous C_3 (46.2° N, 107.4° E)										
M 225–M 233	16–1	Gobi—Mandach	Tuffites	350.6	−10.6	5.1	23.6	35	AF	30 mT
M 234–M 237	16–2	Gobi—Mandach	Tuffites	328.4	9.5	3.6	121.2	14	AF	12 mT
M 238–M 240	16–3	Gobi—Mandach	Tuffites	338.9	−8.5	2.3	454.6	10	AF	20 mT
M 341–M 346	29	Urtuchuraj	Tuffs	332.0	−1.3	8.7	14.9	20	TC	200–500° C
										400–660° C
M 376–M 383	37	Ulan Bator	Siltstones	324.0	−2.5	4.8	53.2	18	AF	40–100 mT
Permian P_1 (47.8° N, 107.1° E)										
M 50–M 56	5	Tel-tin gol	Tuffites	330.2	33.6	4.9	23.1	39	TC	350–500° C
										600–660° C
M 101–M 105	7	Chulut Tsagan Del	Andesites	333.0	18.0	8.4	27.4	12	TC	550–600° C
									AF	40–80 mT
M 84–M 85	9	Chulut Tsagan Del	Sandstones	332.4	16.6	5.6	39.7	18	AF	30 mT
M 250–M 258	17	Modoto	Silicites	313.9	19.5	9.7	15.5	16	TC	500–680° C
Triassic (49.1° N, 105.5° E)										
M 16–M 31	2	Barun Buren	Andesite tuffs	83.7	−39.2	4.5	153.3	8	TC	640° C
M 96–M 100	6	Chulut Tsagan Del	Andesite tuffs	83.0	−34.1	11.3	9.8	19	TC	550° C
M 305–M 309	20	Tel-tin gol	Andesites	117.3	−45.4	3.5	1248.6	3	TC	450–620° C
M 310–M 313	21	Selenga gol	Andesites	304.5	57.1	4.7	79.9	13	TC	450–580° C
M 314–M 317	22	Selenga gol	Andesites	103.0	−69.5	6.1	403.8	6	TC	500–690° C
M 330–M 333	26	Šandin Dava	Andesite tuffs	98.7	−61.0	2.6	206.9	16	TC	300–500° C
Cretaceous (46.1° N, 107.5° E)										
M 86–M 90	12	Chulut Tsagan Del	Basalts	175.8	−62.8	5.6	41.0	17	TC	450° C
M 91–M 95	13	Chulut Tsagan Del	Basalts	167.1	−63.7	6.1	31.0	19	TC	550° C
M 201–M 219	14	Gobi	Sandstones	202.7	−58.7	5.8	21.8	30	TC	450–620° C

Desert—Mandach (Table 1). A criterion of palaeomagnetic stability is the existence of established normal and reverse directions of remanent magnetization (RM) (reverse, R = 45%) within one exposure and one sample in Upper Carboniferous tuff from locality 29 Urtuchuraj after thermal demagnetization (Fig. 2). Silicified siltstones from an exposure in Ulan Bátor, locality 37, yielded reverse RM directions after demagnetization by the AF field in an interval of 40–100 mT.

5. Lower Permian

From the petrophysical as well as palaeomagnetic viewpoint, very interesting data were furnished by silicified andesite tuffs of the Lower Permian from locality 5, Tel-tin gol. Figure 3 shows a typical example from one locality comprising samples reversely polarized (represented by specimen M 51/4), normally polarized (specimen M 55/1) and both normally and reversely polarized (specimen M 56/3). Within the same exposure and specimens of the same sample, groups were singled out that included 56% of reverse RM components with mean direction $D = 163°$, $I = -32°$, and another group (44%) with normal RM direction $D = 314°$, $I = 35°$. The magnetization carriers are magnetite and haematite. Andesites and sandstones from localities 7 and 9, Tshulut Cagan Del, were subjected to demagnetization of thermal and alternating fields with resultant normal RM directions. Permian siltstones from the exposure near Modoto,

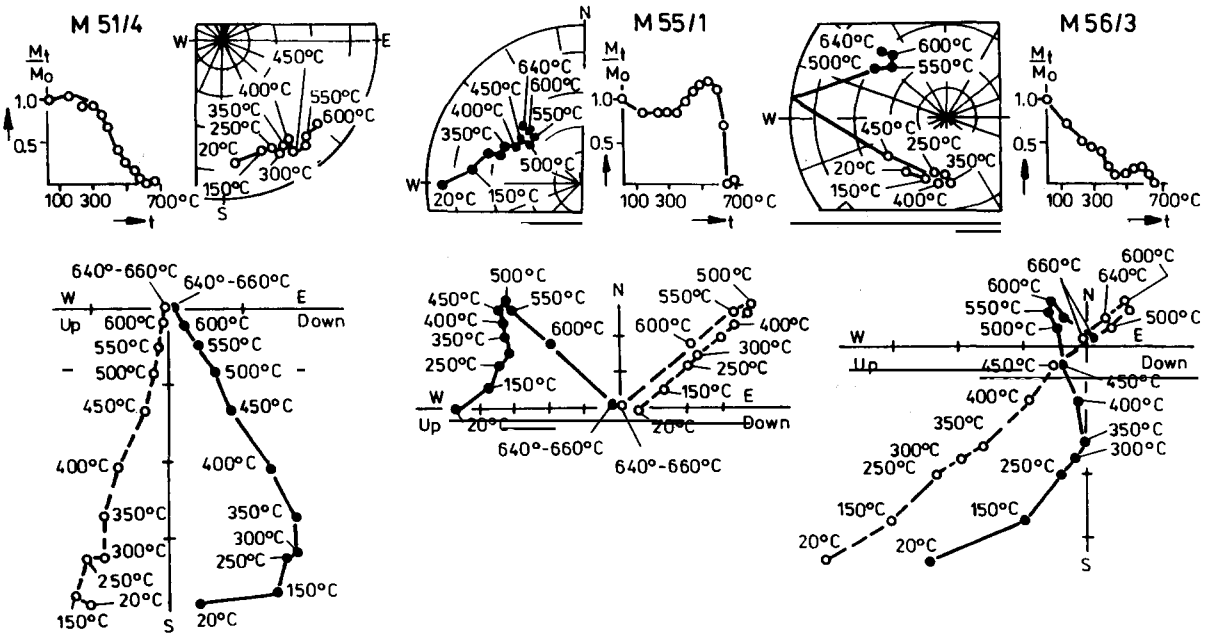


Fig. 3. Lower Permian silicified andesite tuffs, locality Tel-tin gol. The figure shows normalized graphs of M_t/M_0 dependent on temperature and stereographic projection of remanent magnetization on to lower (upper) hemisphere by full (open) circle. In the Zijderveld diagrams, the full (open) circles denote projection on to horizontal (vertical) planes.

locality 17, also displayed normal RM components with the blocking temperature of haematite. The andesite tuffites of north Mongolia (Fig. 4) are a characteristic example of the presence of both the reverse components ($D = 253°$, $I = -72°$) and a strong recent one ($D = 353°$, $I = 60°$). The direction of normally polarized components is close to the recent geomagnetic field. The reverse polarity most probably corresponds to Cretaceous palaeomagnetic field (Fig. 5).

6. Triassic

Andesite tuffs from locality 2, Barun Buren, comprise a group with stable recent normal RM direction ($D = 349°$, $I = 59°$) on the one hand, and on the other, another group with a strong reverse component of remanent magnetization ($D = 84°$, $I = -39°$). Triassic andesites from localities 20, 21 and 22, Tel-tin gol and Selenga gol, exhibit the presence of both normal directions of remanent magnetization with the blocking temperature of haematite and of stable reverse components with haematite as the magnetization carrier. Andesite tuffs from locality 26, Shandin Dava, and locality 6, Tshulut Cagan, also exhibited very stable reverse RM components. In both cases haematite is the magnetization carrier. Of the total number of Triassic samples interpreted by normal and reverse, N = 23% and R = 77%.

7. Cretaceous

Samples of Cretaceous basalts from two localities 12 and 13 in central Mongolia yielded palaeomagnetic directions, documented by multicomponents analysis, Fisher's (1953) statistics, and the ability not to form phase changes in the course of thermal demagnetization up to the Curie temperature of magnetite. Both groups have reverse components of RM. Cretaceous sandstones from the Gobi Desert, locality 14, exhibit a strong recent normal RM component ($D = 0°$, $I = 72°$) beside a stable reverse ($D = 203°$, $I = -59°$) in the temperature interval of 450–620 °C.

8. Palaeogeographic interpretation and conclusions

The preliminary data (Pruner, 1987) are extended by this study which includes Carboniferous and Permian rocks as well as Triassic rocks. The groups of samples 2 (M16–M31) and 6 (M96–M100) originally assigned to the Upper Permian (P3) are according to recent geological investigation assigned to the Triassic. The positions of new palaeomagnetic poles for the Carboniferous to Cretaceous differ only slightly from those in the previous work (poles given in brackets): Carboniferous = 38 ° N, 320 ° E (41 ° N, 307 ° E); Permian P_1 = 45 ° N, 335 ° E (45 ° N, 337 ° E); Cretaceous = 87 ° N, 253 ° E (83 ° N, 220 ° E). The conclusions about the palaeogeographical development of Mongolia with respect to the Siberian platform during the Carboniferous and Permian have been confirmed in the present study; similar apparent polar wander paths were derived both for Mongolia and the north China block.

The palaeomagnetic and palaeogeographic data in Table 2 are interpreted in relation to the Siberian platform and the north China block. The

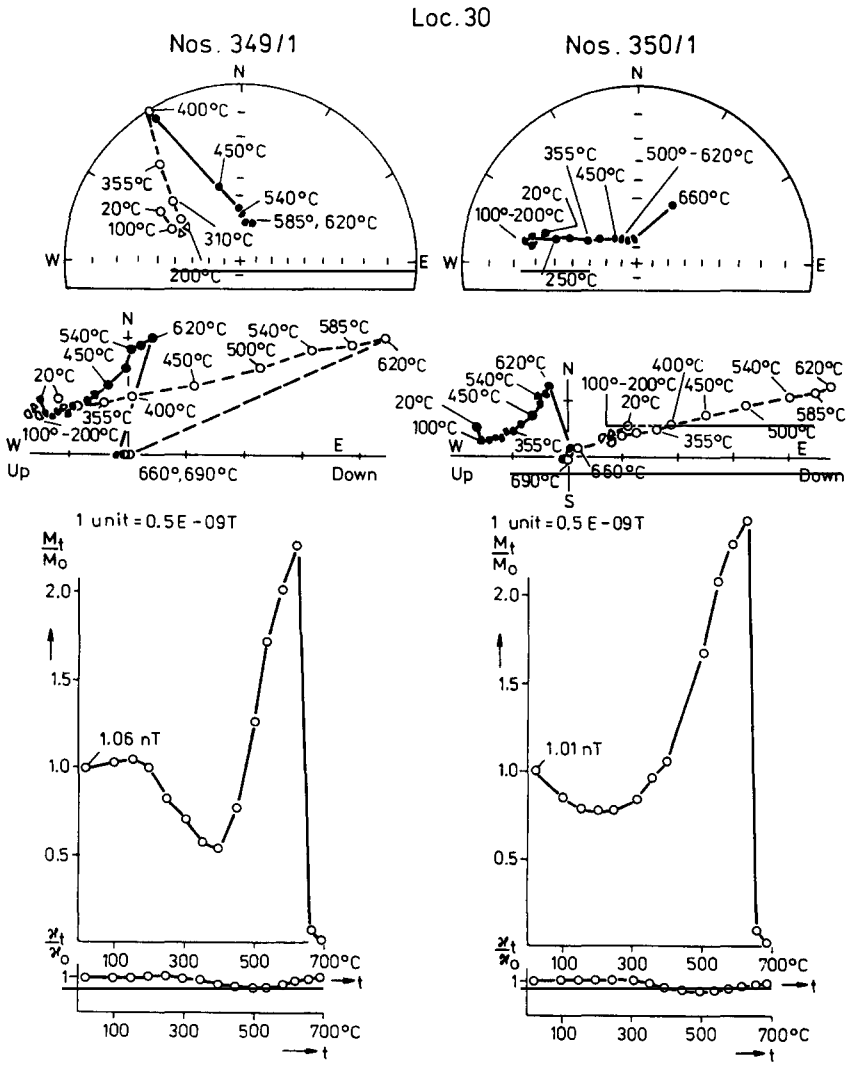


Fig. 4. Thermal demagnetization of Permian andesite tuffites of north Mongolia. For explanations see Fig. 2.

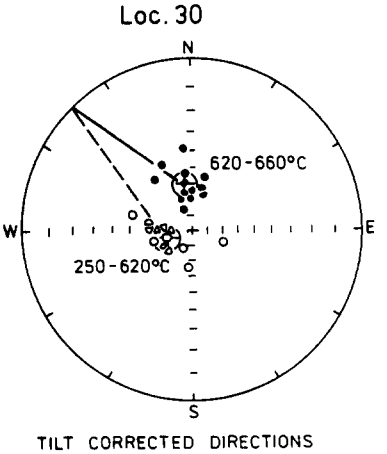


Fig. 5. Permian andesite tuffites, locality 30. The specimens show two magnetization components, reverse magnetization corresponding most probably to the Cretaceous and normal magnetization to the recent geomagnetic field. The mean tilt of the rocks is 39° SE.

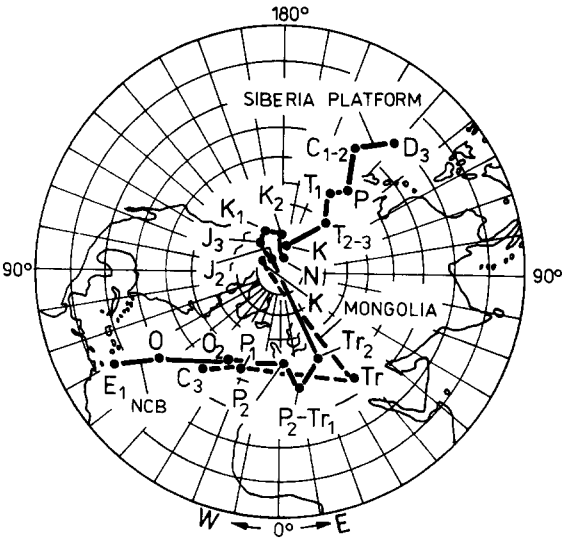


Fig. 6. Palaeomagnetic pole positions for: the Upper Carboniferous, C3; Lower Permian, P_1; Triassic and Cretaceous of Mongolia. Apparent polar wander paths for north China block (NCB) (Lin and Fuller, 1990) and Siberia platform (Khramov, 1982).

positions of palaeomagnetic poles for Mongolia and the Siberian platform differ markedly, and the apparent polar wander paths are also different (Fig. 6). The positions and apparent polar wander paths are almost coincident for Mongolia and the north China block.

From the palaeomagnetic data for the Carboniferous to the Cretaceous of Mongolia we derived values of palaeolatitudes and then correlated them with the Siberian platform (Irving et al., 1976; Khramov, 1982; Krs, 1982), north China (McElhinny et al., 1981; Lin and Fuller, 1990), Tarim (Bai et al., 1987) and Altay–Sayan blocks (Kravchinskij, 1979; Khramov, 1982) (Figs. 7–10).

On the basis of palaeogeographic parameters it may be inferred that in the geological past the region of Mongolia was part of neither the Siberian platform nor Eurasia. In the Cretaceous there are differences in palaeorotation and palae-

TABLE 2

Mean palaeomagnetic directions and pole positions for Carboniferous, Permian, Triassic and Cretaceous rocks in Mongolia

Stratigraphy group no.	Mean palaeomagnetic directions		α 95	K	n	N	Percentage of polarity		Palaeomagnetic pole positions		α 95
	D(°)	I(°)					N(%)	R(%)	Latitude (°N)	Longitude (°E)	
Carboniferous C_3 16-1; 2; 3; 29; 37	334.7	−2.7	12.4	39.2	97	5	91	9	37.5	320.1	10.4
Permian P_1 5; 7; 9; 17	327.3	22.1	13.2	49.3	85	4	74	26	44.8	335.1	11.6
Triassic 2; 6; 20; 21; 22; 26	99.7	−52.2	14.8	21.4	65	6	23	77	32.0	32.7	16.8
Cretaceous 12; 13; 14	182.9	−62.6	14.2	76.8	66	3	–	100	86.9	252.8	21.4

n, number of samples; N, number of localities.

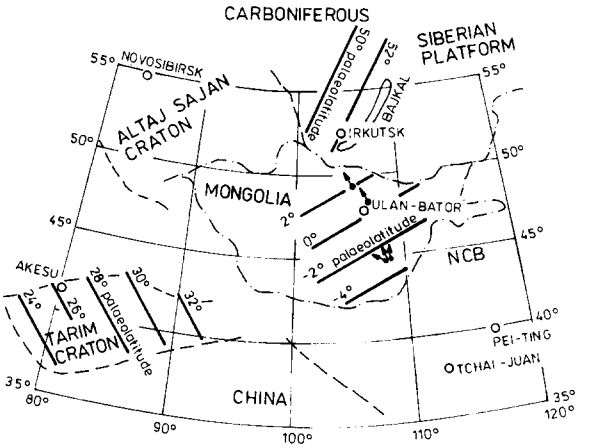


Fig. 7. Upper Carboniferous palaeomagnetic declinations and palaeolatitudes for Mongolia; Carboniferous palaeolatitudes for the central Siberia and the Tarim cratons.

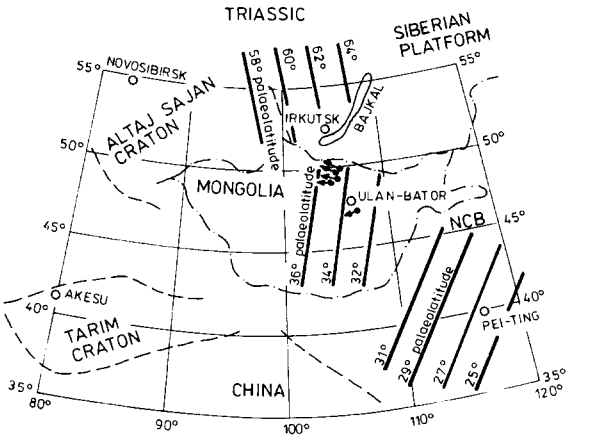


Fig. 9. Triassic palaeomagnetic declinations and palaeolatitudes for Mongolia; Triassic palaeolatitudes for the central Siberia and north China (NCB) blocks.

olatitudes of about 15°. This is presumably the period in which the ocean basin between the Siberian platform and Mongolia was closing. A coincident orientation of palaeomeridians and palaeolatitude values can be observed in the Triassic for the areas of Mongolia and the north China block; the Siberian platform and the Altay–Sayan block were situated 30° or even further north than the area of Mongolia.

For the Permian, differences in palaeorotation and palaeolatitudes are again very small (about 15°) between the north China block and Mongolia. However, between the Siberian platform and Mongolia, differences in palaeorotation amount to as much as 150°, and in palaeolatitudes to as much as 50°. In the Permian and the Carboniferous, the palaeogeographic position of the Tarim block was different from that of the Siberian platform, Mongolia and the north China block. In the Carboniferous, Mongolia was situated in the equatorial zone.

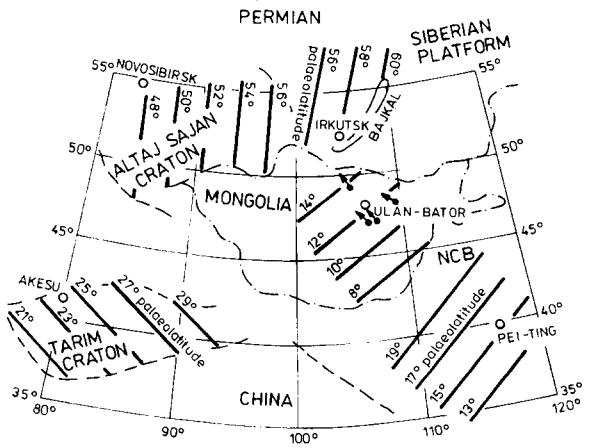


Fig. 8. Lower Permian palaeomagnetic declinations and palaeolatitudes for Mongolia; Permian palaeolatitudes for the central Siberia, Altay–Sayan, Tarim and north China (NCB) blocks.

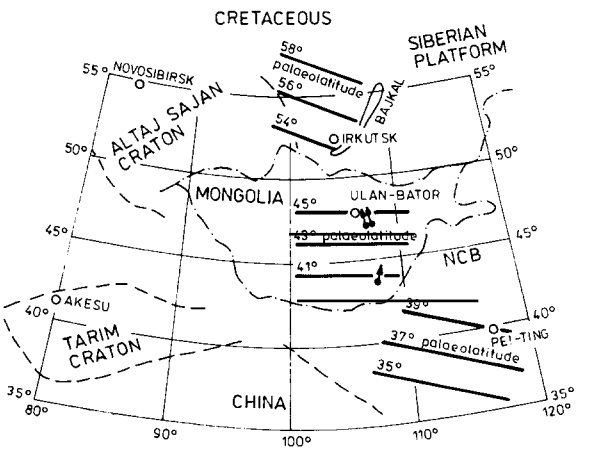


Fig. 10. Cretaceous palaeomagnetic declinations and palaeolatitudes for Mongolia; Cretaceous palaeolatitudes for the central Sibera and north China (NCB) blocks.

The above palaeogeographic data, which have been correlated with those of some blocks of Eurasia and other continents make it evident that in the Late Palaeozoic and the Mesozoic the area of Mongolia and north China formed part of a single block. We may also presume that the ocean basin between the Siberian platform and Mongolia did not begin closing until the Cretaceous. The finding that the area of Mongolia relative to the Siberian platform had a different palaeogeographic position from the Carboniferous until the Triassic, is based on differences in palaeolatitudes (as much as 50°) and palaeorotation values, which attain almost 150°. These results confirm the interpretation, made on the basis of similar positions of palaeomagnetic poles and an almost identical pole wander path, that the part of Mongolia under investigation, together with the north China block, were not part of Eurasia from the Carboniferous until the Triassic, but of the Pacific Plate.

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